Sri Lanka Institute of Information Technology

Assignment Part 2. A React Frontend Application

Application Frameworks

(SE3040)

SE_26

AGRIHUB-LK – Deployment Report

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Deployment Date: 4/12/2026

Repository Link: https://github.com/SandeepaChathumina/AgriHUB-LK

1. Deployment Overview

The AgriHUB-LK application is split into two independently deployed services:

Layer	Platform	Technology
Frontend	Vercel	React 19 + Vite
Backend	Railway	Node.js + Express 5
Database	MongoDB Atlas	Mongoose 9

2. Frontend Deployment – Vercel

Platform: Vercel (https://vercel.com)

2.1 Setup Steps

1. Log in to Vercel and click "Add New Project."
2. Import the GitHub repository.
3. Set the Root Directory to frontend/.
4. Vercel will automatically detect Vite. Confirm the following build settings:
   - Framework Preset: Vite
   - Build Command: npm run build
   - Output Directory: dist
5. Add the required environment variable under Environment Variables (refer to Table 2.2).
6. Click Deploy. Vercel will install dependencies and build the application.
7. Verify that the deployed URL loads the application correctly.

2.2 Environment Variables (Vercel Project Settings)

Variable	Value
VITE_API_BASE_URL	Railway backend URL
Port	5173

2.3 Deployed URL

https://agri-hub-lk-frontend.vercel.app/

3. Backend Deployment - Railway

Platform: Railway (https://railway.app)

3.1 Setup Steps

1. Log in to Railway and click “New Project → Deploy from GitHub repo.”
2. Select the repository and set the Root Directory to backend/.
3. Railway will automatically detect Node.js. Set the start command to:
   node index.js
4. Add all required environment variables under the Variables tab (refer to Table 3.2).
5. Click Deploy. Railway will run npm install and start the server.
6. Verify that the service is live by calling the AgrihubLK API Server check endpoint:

   GET https://<railway-url>/

   Expected Response:

```
{
  "success": true,
  "message": "AgrihubLK API Server"
}
```

3.2 Environment Variables (Railway Project Settings)

Variable	Value
PORT	3000
MONGO_URI	your_mongodb_connection_string
JWT_SECRET	your_jwt_secret
EMAIL_USER	your_email_address
EMAIL_PASS	your_email_app_password
EXCHANGE_RATE_API_KEY	your_exchange_rate_api_key
STRIPE_SECRET_KEY	your_stripe_secret_key
STRIPE_PUBLISHABLE_KEY	your_stripe_publishable_key
GEMINI_API_KEY	your_gemini_api_key
MESSAGE_SECRET_KEY	your_message_secret
CLOUDINARY_CLOUD_NAME	your_cloudinary_cloud_name
CLOUDINARY_API_KEY	your_cloudinary_api_key
CLOUDINARY_API_SECRET	your_cloudinary_api_secret
CLIENT_URL	http://localhost:5173

3.3 Deployed URL (Fill in after deployment)

https://agrihub-lk-production.up.railway.app

4. Database - MongoDB Atlas

The database is hosted on MongoDB Atlas. No direct deployment action is required for the database itself. The Railway backend connects to the database using the MONGODB_URI environment variable.

Ensure that the MongoDB Atlas cluster's Network Access configuration allows connections from all IP addresses by adding:

0.0.0.0/0

This is necessary because Railway uses dynamic IP addresses which may otherwise be blocked.

5. CORS Configuration

The backend restricts cross-origin requests to the origin specified in the CLIENT_URL environment variable.

This value must exactly match the Vercel deployment URL (without a trailing slash) to ensure proper communication between the frontend and backend services.

6. Deployment Checklist

- MongoDB Atlas network access allows all IPs (0.0.0.0/0)
- All backend environment variables are configured in Railway
- CLIENT_URL in Railway is set to the Vercel frontend URL
- VITE_API_BASE_URL in Vercel is set to the Railway backend URL
- Backend AgrihubLK API Server check verified: GET / returns { "success": true }
- Frontend production build succeeds: npm run build in frontend/
- Authentication flow tested on deployed URLs (login sets HttpOnly cookie correctly)